

Module2

Chapter 1

Introduction to Linear Algebra

Linear equations and systems

Let's solve a simple puzzle:

There are 27 pieces of fruit in a barrel, and twice as many oranges as apples. How many apples and oranges are in the barrel?

Can we rewrite this question mathematically?

Let x be the number of apples and y be the number of oranges:

Then,

$$x + y = 27$$

$$y = 2x$$

$$x^2 + 2y + z = 3 \quad \times$$

$$\frac{1}{x} + 2y + z = 0 \quad \times$$

$$\sqrt{x} + y = 0 \quad \times$$

Let's decipher the above representation of the problem:

What we see above is an example of a **linear system**.

Linear system is a collection of equations in which variables are multiplied by constants and summed, with the following conditions: *There are no powers of variables greater than one, no fractional or negative powers of variables, and no places where the variables are multiplied together. Such equations are called*

Linear equations. These equations represent *straight lines* in geometry, hence the term Linear.

The general linear equation is represented as $\underline{u_1x_1} + \underline{u_2x_2} + \dots + \underline{u_nx_n} = \underline{v}$

Here,

x

- u 's – represent the coefficients
- x 's – represent the unknowns
- v – represents the constant

Gauss Jordan Elimination

Let's revisit our linear system:

$$x + y = 27$$

$$x + 2x = 27$$

$$y = 2x$$

$$3x = 27$$

$$\underline{x = 9}$$

$$y = 2x \rightarrow y = 9 \times 2 = \underline{18}$$

A simple way of solving will be *method of substitution*. Where we represent one variable in terms of the other, using one of the equations and replace it's value in the other equation. So, here, from the second equation, we get $y = 2x$. We then replace y by $2x$ in the first equation and solve for x . Our first equation now becomes:

$$x + 2x = 27$$

This gives us $3x = 27$. Thus, $x = 9$. We use this value of x to calculate the value of y . We know $y = 2x$. Thus, $y = 2 \times 9 = 18$. Thus, we have 9 apples and 18 oranges!

What happens if, instead of 2 unknowns, we had 10 unknowns and 10 linear equations in a system?

How about 100 unknowns and 100 equations?

One can see, as the number of variables/unknowns increase, method of substitution becomes very cumbersome. We need to use a better way to solve linear systems!

Gauss Jordan elimination is often preferred over method of substitution when dealing with large systems of linear equations since it provides a systematic way to solve the problem, making it easier to manage complex systems with many variables.

This chapter is dedicated to learning Gauss Jordan Elimination.

Our first step is to rewrite the linear equations in “***general linear equation***” form:

$$x + y = 27$$

$$2x - y = 0$$

The next step involves writing the above system using *matrices* and *vectors*:

$$\begin{array}{c} \text{Coefficient} \\ \text{Matrix} \end{array} \leftarrow \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{array}{c} \text{Variable} \\ \text{vector} \end{array} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 27 \\ 0 \end{bmatrix} \rightarrow \begin{array}{c} \text{Constant} \\ \text{vector} \end{array}$$

What we see here is an example of *Matrices and Vectors*.

Matrices are collection of objects arranged rows and columns.

Vectors are Matrices containing one row or one column.

Going back to our example, we solved it using method of substitution and got:

$$x = 9$$

$$y = 18$$

We can rewrite the above as:

$$x + 0y = 9$$

$$0x + y = 18$$

We can thus, represent our solution using vectors and matrices as:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 18 \end{bmatrix}$$

We just need to figure out a way to start from:

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 27 \\ 0 \end{bmatrix} \text{ and reach } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 18 \end{bmatrix}.$$

The method we use is called *Gauss Jordan Elimination*.

*The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is called an Identity Matrix. A Matrix that has all 1's along it's Diagonal and 0's elsewhere is called an **Identity Matrix**.*

Our first step in Gauss Jordan elimination is to write the system of equations in Matrix form (which we already did). Then we rewrite the Matrices in **Augmented matrix Form**. So, our Linear system becomes:

$$\left[\begin{array}{cc|c} 1 & 1 & 27 \\ 2 & -1 & 0 \end{array} \right] \text{ and ideally our solution system should look like } \left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & 18 \end{array} \right]$$

Gauss Jordan elimination gives us a bunch of moves that we can apply to the initial augmented matrix for the linear system to convert it to the solution Matrix. *The idea is to keep working on the augmented matrix till we get an identity matrix on the left side of the vertical line in the Augmented Matrix. Once we reach there, whatever we have on the right of the vertical line, is our solution.*

The three valid moves we can make are:

- **Multiplication/Division by a scalar** – Multiply a row by a non-zero scalar
- **Row combination** - Replace a row with the sum/difference of that row and a (nonzero) multiple of another row.
- **Swapping** - Swap two rows.

Let's try applying the above moves to the original augmented matrix!

$$\begin{bmatrix} 1 & 1 & | & 27 \\ 2 & -1 & | & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & | & - \\ 0 & 1 & | & - \end{bmatrix}$$

Add Row2 to Row1 ($R1 = R1 + R2$)

$$\begin{bmatrix} 3 & 0 & | & 27 \\ 2 & -1 & | & 0 \end{bmatrix}$$

Multiply Row1 by $1/3$ ($R1 = R1/3$)

$$\begin{bmatrix} 1 & 0 & | & 9 \\ 2 & -1 & | & 0 \end{bmatrix}$$

$$\begin{aligned} 2 - 2 \times 1 &= 0 \\ -1 - 0 \times 2 &= -1 \\ 0 - 2 \times 9 &= -18 \end{aligned}$$

Subtract 2Row1 from Row2 ($R2 = R2 - 2R1$)

$$\begin{bmatrix} 1 & 0 & | & 9 \\ 0 & -1 & | & -18 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiply Row2 by -1 ($R2 = -1R2$)

$$\left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & 18 \end{array} \right]$$

$$6/12 \quad 1/2$$

Hence, our solution is: $x = 9$ and $y = 18$!

Something important to remember is:

The number of equations in the linear system is the number of rows r in the augmented matrix, and the number of columns k in the matrix left of the vertical line is the number of unknowns.

The result of Gauss Jordan Elimination is always a matrix in RREF!

Reduced Row Echelon Form (RREF)

A matrix is said to be in RREF if it has the following properties:

1. The first non-zero element in each row is called the *Pivot* of that row. In RREF. The pivot of each row is 1.
2. The pivot of any given row is always to the right of the pivot of the row above it.
3. The pivot is the only non-zero entry in its column.

Is an Identity Matrix, also called I , in RREF?

When we can reach one matrix from the other using Gauss Jordan Elimination or if they can be reduced to the same RREF Matrix, they are called Row Equivalent Matrices.

Let's Practice:

Find the solution to the following system of equations:

$$x + y - z = 0$$

$$3x - 2y + 2z = 5$$

$$5x - y - z = 0$$

Is the following matrix in RREF?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ Yes}$$

Solution set of a system of linear equations:

It isn't always the cases that a system of equations will have a unique solution. In fact, there are three possible cases:

- There are no solutions, because the equations are not logically consistent with each other. Let's say we have two equations in a system:

$$x + y = 1$$

$$x + y = 0.$$

These statements can't both be true. (Note: Equations may contradict each other in less obvious ways.) Such systems are called *inconsistent systems*.

If Gauss-Jordan elimination results in a row of all zeros, except for the number on the right-hand side of the augmented matrix, then there are no solutions:

$$x + y - z = 0$$

$$3x - 2y + 2z = 5$$

$$5x - y - z = 0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 1: Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 3 & -2 & 2 & 5 \\ 5 & -1 & -1 & 0 \end{array} \right]$$

Step 2: Gauss-Jordan Elimination

$$R_3 = R_3 - 5R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 3 & -2 & 2 & 5 \\ 0 & -6 & 4 & 0 \end{array} \right]$$

$$R_2 = R_2 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -5 & 5 & 5 \\ 0 & -6 & 4 & 0 \end{array} \right]$$

$$R_2 = R_2 / 5$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & -6 & 4 & 0 \end{array} \right]$$

$$R_3 = R_3 / 2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & -3 & 2 & 0 \end{array} \right]$$

$$R_1 = R_1 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 \\ 0 & -3 & 2 & 0 \end{array} \right]$$

$$R_2 = -1R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & -3 & 2 & 0 \end{array} \right]$$

$$R_3 = R_3 + 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & -3 \end{array} \right]$$

$$R_3 = -1R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$R_2 = R_2 + R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$\therefore x = 1, y = 2, z = 3$$

$$\left[\begin{array}{cccc} \textcircled{1} & 0 & 0 & 0 \\ 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & \textcircled{1} \end{array} \right]$$

$\therefore$ YES

All pivots are 1

All pivots are to the right of the pivot of the Row above

Pivots are the only non-zero entry in their column

Let's Practice:

Find the solution to the following system of equations:

$$x + y - z = 0$$

$$3x - 2y + 2z = 5$$

$$5x - y - z = 0$$

Is the following matrix in RREF?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ Yes}$$

Solution set of a system of linear equations:

It isn't always the cases that a system of equations will have a unique solution. In fact, there are three possible cases:

- There are no solutions, because the equations are not logically consistent with each other. Let's say we have two equations in a system:

$$x + y = 1$$

$$x + y = 0.$$

These statements can't both be true. (Note: Equations may contradict each other in less obvious ways.) Such systems are called *inconsistent systems*.

If Gauss-Jordan elimination results in a row of all zeros, except for the number on the right-hand side of the augmented matrix, then there are no solutions:

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 1 & 0 \end{array} \right]$$

If we apply Gauss-Jordan elimination to this augmented matrix, we get:

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 1 & 0 \end{array} \right]$$

$$R1 = R1 - R2$$

$$\left[\begin{array}{cc|c} 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \right]$$

We have a row of all 0's on one side and a non-zero number on the other side in the Augmented matrix.

- There are an infinite number of solutions. There are times when we have more than 1 equations giving the same information. Hence, it's as good as having one equation. In general, if the number of unique equations is less than the number of unknowns, it leads to infinite number of solutions since there are not enough constraints. For example, if we have a system with the following equations:

$$x + y = 1$$

$$2x + 2y = 2.$$

Normally two equations describe lines that intersect at a point, but here, they really describe the same line.

If Gauss Jordan Elimination leads to a row of all 0's on either side of the Augmented Matrix and the number of remaining non zero rows is less than the number of unknowns, then it's the case of infinite solutions. Such systems are called *dependent systems*.

$$\begin{bmatrix} 1 & 1 & | & 1 \\ 2 & 2 & | & 2 \end{bmatrix}$$

Applying Gauss Jordan Elimination we get:

$$\begin{bmatrix} 1 & 1 & | & 1 \\ 2 & 2 & | & 2 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & | & 1 \\ 0 & 0 & | & 0 \end{bmatrix}$$

We have a row of all 0s on either side of the Augmented Matrix and the number of non-zero rows is 1 which is less than the number of unknowns which is 2.

- If you reach RREF where you have one row $\rightarrow [0 \ 0 \ \dots \ | \ 0]$ and another as $[0 \ 0 \ \dots \ | \ \text{NonZero}]$ then it's **Inconsistent**
- The third case is the one in our previous examples: exactly one solution.

When Gauss-Jordan elimination results in ones along the diagonal on the left-hand side, one per variable, and zeros elsewhere on that side (Identity Matrix), we have the case of unique solutions.

For example:

$$\begin{bmatrix} 1 & 1 & | & 27 \\ 2 & -1 & | & 0 \end{bmatrix}$$

Leads to

$$\left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & 18 \end{array} \right]$$

A square matrix that leads to a unique solution is called a Non-Singular matrix while the ones that do not lead to unique solutions are called Singular Matrices.

(We cover more about singular and non-singular matrices in the subsequent chapters!)

Let's Practice:

Reduce each of the following Matrices to RREF:

1. $\begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 0 & -1 \\ -1 & -2 \end{bmatrix}$

2. $\begin{bmatrix} -1 & 1 \\ -1 & 2 \\ -3 & 2 \end{bmatrix}$

Solve the following system of equations:

$$x + y + z = 2$$

$$x + 2y + 3z = 5$$

$$2x + 3y + 4z = 11$$

$$1. \begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 0 & -1 \\ -1 & -2 \end{bmatrix}$$

Apply Gauss Jordan

$$R_1 = -1 R_1$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 0 \\ 0 & -1 \\ -1 & -2 \end{bmatrix}$$

$$R_2 = R_2 + R_1$$

$$R_4 = R_4 + R_1$$

$$\begin{bmatrix} 1 & -1 \\ 0 & -1 \\ 0 & -1 \\ 0 & -3 \end{bmatrix}$$

$$R_2 = -1 R_2$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 0 & -1 \\ 0 & -3 \end{bmatrix}$$

$$R_1 = R_1 + R_2$$

$$R_3 = R_3 + R_2$$

$$R_4 = R_4 + 3 R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$2. \begin{bmatrix} -1 & 1 \\ -1 & 2 \\ -3 & 2 \end{bmatrix}$$

Apply Gauss Jordan

$$R_1 = -R_1$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 2 \\ -3 & 2 \end{bmatrix}$$

$$R_2 = R_2 + R_1$$

$$R_3 = R_3 + 3 R_1$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}$$

$$R_1 = R_1 + R_2$$

$$R_3 = R_3 + R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$x + y + z = 2$$

$$x + 2y + 3z = 5$$

$$2x + 3y + 4z = 11$$

Write augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 3 & 4 & 11 \end{array} \right]$$

Apply Gauss Jordan Elimination

$$R_2 = R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 2 & 3 & 4 & 11 \end{array} \right]$$

$$R_3 = R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 7 \end{array} \right]$$

$$R_3 = R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 4 \end{array} \right]$$

→ Inconsistent System
∴ No solutions

Extra Practice (It's fun)

Solve using Gauss Jordan Elimination :

1. $y - 8z = 9$

$$x - 2y + 3z = -3$$

$$7y - 5z = 12$$

2. $x + 2y - 3z = -2$

$$3x - y - 2z = 1$$

$$2x + 3y - 5z = -3$$

3. $3x - 2y + z = -6$

$$4x - 3y + 3z = 7$$

$$2x + y - z = -9$$

4. Two numbers add to 7. Their difference (first minus second) is 1. Find the numbers.

1.

$$\left[\begin{array}{ccc|c} 0 & 1 & -8 & 9 \\ 1 & -2 & 3 & -3 \\ 0 & 7 & -5 & 12 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & -3 \\ 0 & 1 & -8 & 9 \\ 0 & 7 & -5 & 12 \end{array} \right] \xrightarrow{R_3 = R_3 - 7R_2}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & -3 \\ 0 & 1 & -8 & 9 \\ 0 & 0 & 51 & -51 \end{array} \right] \xrightarrow{R_3 = R_3 / 51} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -3 \\ 0 & 1 & -8 & 9 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$$\xrightarrow{R_2 = R_2 + 8R_3} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right] \xrightarrow{R_1 = R_1 - 3R_3}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right] \xrightarrow{R_1 = R_1 + 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$$\therefore x = 2$$

$$y = 1$$

$$z = -1$$

$$2. \left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 3 & -1 & -2 & 1 \\ 2 & 3 & -5 & -3 \end{array} \right] \xrightarrow{R_2 = R_2 - 3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & -7 & 7 & 7 \\ 2 & 3 & -5 & -3 \end{array} \right]$$

$$\xrightarrow{R_2 = R_2 / 7} \left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & -1 & 1 & 1 \\ 2 & 3 & -5 & -3 \end{array} \right] \xrightarrow{R_3 = R_3 - 2R_1}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 1 & 1 \end{array} \right] \xrightarrow{R_2 = R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 2 & -3 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 \end{array} \right]$$

Infinite solutions!

$$3. \left[\begin{array}{ccc|c} 3 & -2 & 1 & -6 \\ 4 & -3 & 3 & 7 \\ 2 & 1 & -1 & -9 \end{array} \right] \xrightarrow{\substack{R_1 = -R_1 + R_2 \\ R_3 = 2R_3 - R_2}} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 13 \\ 4 & -3 & 3 & 7 \\ 0 & 5 & -5 & -25 \end{array} \right]$$

$$\xrightarrow{\substack{R_2 = R_2 - 4R_1 / R_2 = -R_2 \\ R_3 = R_3 / 5}} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 13 \\ 0 & -1 & 5 & 45 \\ 0 & 1 & -1 & -5 \end{array} \right] \xrightarrow{\substack{R_1 = R_1 + R_3 \\ R_3 = R_3 + R_2}}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 8 \\ 0 & -1 & 5 & 45 \\ 0 & 0 & 4 & 40 \end{array} \right] \xrightarrow{\substack{R_2 = -1R_2 \\ R_3 = R_3 / 4}}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 8 \\ 0 & 1 & -5 & -45 \\ 0 & 0 & 1 & 10 \end{array} \right] \xrightarrow{\substack{R_1 = R_1 - R_3 \\ R_2 = R_2 + 5R_3}}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 10 \end{array} \right]$$

$$\therefore \begin{aligned} x &= -2 \\ y &= 5 \\ z &= 10 \end{aligned}$$

4.

$$x + y = 7$$

$$x - y = 1$$

augmented matrix :

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 1 & -1 & 1 \end{array} \right]$$

$$\downarrow R_2 = R_2 - R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & -2 & -6 \end{array} \right]$$

$$\downarrow R_2 = R_2 / -2$$

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & 1 & 3 \end{array} \right]$$

$$\downarrow R_1 = R_1 - R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 3 \end{array} \right]$$

$$\Rightarrow \begin{array}{l} x = 4 \\ y = 3 \end{array}$$